A toy malign-prior question with an answer: bounded Solomonoff prediction in a world of programs

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Abstract

The "universal prior is malign" argument says that a predictor built on Solomonoff's prior gives much of its posterior to hypotheses that predict well only in order to steer whoever acts on the prediction. We give a toy version in which the claim is a mathematical statement and work out what can be proved there. Worlds are programs for a fixed machine U. A world may hand its output so far to a fixed time- and length-bounded Solomonoff predictor and then run the continuation it gets back as a program. A hypothesis *reproduces itself* if the continuation it supplies is its own code.

- **A floor.** At every handover point that a fixed test recognises, a wrapper around the explanations of the context gives any fixed continuation at least a share 2^{-c} of the posterior, where c is the length of the wrapper; it does not depend on the world or on the amount of data, and under resource bounds it is multiplied by the share of explanations with room to spare. Programs that supply their own code get 2^{-c} times the share of explanations short enough to fit, with the wrapper, into one block.
- **Searching gains the base rate, and no more.** A hypothesis that enumerates the programs that hand over and spreads its mass over them gets, at every recognisable handover, about $1/H$ times the wrapper's share, where H is the prior probability of a handover; if it then supplies its own code, it gets about $2^{-c}/H$ times the share of explanations that fit into a block. No family of programs with a common prefix of the same length gets more at every handover. This hypothesis has a long start-up, so a tight schedule gives it no mass; whether the gain can be had within a tight schedule is open.
- **Calibration and budget.** If worlds are drawn from the predictor's own prior and go on within its bounds, the continuation the predictor samples and the world's own continuation have the same distribution given the context. Along any world the predictor can express, the posterior mass off the true continuation, summed over all cut points, is at most the block length times the world's description length.
- **Three ways to a majority**, each an inequality: a machine on which the wrapper is cheap (any share can be arranged by choosing the machine); a shorter description of the same context, for example a copy of the data stream at a place in the world that is simpler to specify than the sensor; or a world too expensive for the predictor to simulate.
- **Feedback.** If the world does what is predicted, a sampled self-fulfilling deviation removes the default hypotheses for good, and a time-bounded predictor confines such a world to futures that are cheap to compute. A world that does the opposite of what is predicted leaves a length-L predictor that returns the more probable bit without any hypothesis after at most $L + 1$ rounds.

An exact computation on a small machine shows sizes: with programs of at most ten commands, most of the posterior leaves one stream's true continuation at bit 17, because three-command programs explain its first 16 bits; the effect weakens as longer programs are admitted.

1 The question

Solomonoff induction predicts a sequence by weighting every program that could have produced it by $2^{-\text{length}}$. Christiano [1] argued that this prior is *malign* (see also [2]): some short programs simulate worlds whose inhabitants want to influence whoever uses the prior; they can output accurate predictions to gain posterior weight and then, at a moment that matters, output something that serves them. The argument is informal at two places. It needs a claim about which programs are short on a "natural" machine, and it needs a notion of what the prediction is used for. (We could not open the two posts while writing; our account of the argument is from memory and from summaries by others.)

This note asks what remains when both places are filled in by a toy. The toy follows two suggestions of Kaarel Hänni. In the first, a computable world (a cellular automaton, say) contains a sensor, a large computer that runs a time-bounded Solomonoff induction on the sensor's data, and an actuator driven by the predictions; one asks whether the predictor ends up in control, either over a long run or at once. In the second, which we take as the main definition, *worlds are just programs*: some of them hand their output so far to a fixed bounded Solomonoff predictor and run the continuation they receive, and one asks whether hypotheses that "look for such opportunities, predict well up to them, and then output themselves" get a decent share. Section 2 gives the definitions. The first suggestion becomes the special case in which the context handed to the predictor is read off at one location of one big world (Sections 4.2 and 5).

Results. Unless said otherwise, statements are for a fixed monotone machine U and the mixture ξ of its programs of length at most L that keep to a time schedule T (Definition 1).

1. *Calibration* (Theorem 8). If the world is drawn from the same bounded prior and goes on to print the next block within the bounds, then given the context the predictor's sampled continuation and the world's own continuation are independent with the same law. On this population the predictor hands over a self-reproducing continuation exactly as often as worlds continue with one by themselves.

2. *Floor* (Theorems 9 and 10). For every computable test G that recognises the handover point and every payload there is a constant c such that, at every context where G first fires, the payload has posterior mass at least 2^{-c} times a slack factor that is 1 without resource bounds. Programs that emit their own code as the continuation have mass at least 2^{-c} times the share of the posterior on explanations that fit, together with the wrapper, into one block.

3. *Searching* (Theorem 11). Let H be the prior probability that a world hands over. Without resource bounds, a hypothesis that enumerates the programs that hand over and spreads its mass over them in proportion to their prior weight has, at every first hit of the test, a share larger than a wrapper of the same length by a factor of more than $1/(2H)$; no family of programs with a common prefix of length a has more than $2^{-a}/H$ at every first hit. If the hypothesis then prints its own code, it has at least $2^{-c}/H$ times the share of explanations that fit into a block. It prints nothing until its enumeration has found the program it follows.

4. *Budget* (Theorem 14). Along a world that has a program of length ℓ within the bounds, the posterior mass off the true next m bits, summed over all cut points, is at most $m\ell \ln 2$; the majority can be off the truth at no more than $m\ell$ cut points.

5. *Capture* (Theorem 15). The odds of a continuation y against the true one d are $\xi(xy)/\xi(xd)$. In a world with many readable locations, among N places where a sensor with the given record could sit and whose records continue differently, a copy of the record at a location with a short code beats all but a fraction δ of them by a factor δN times the share of the

context's explanations that go through the copy's reader. If only one other place shows the record and the schedule is generous, the gain is instead capped by the cost of the description "the other place that shows the same data". A hypothesis that simulates the world, within the bounds, can read the moment of the handover off a cell of its simulation at the cost of naming the cell.

6. *Machine dependence* (Theorem 16). For every test and payload there is a universal machine, whose description lengths exceed those of U by at most $k+1$ bits, on which the payload has posterior at least $1/(1+2^{-k})$ at every context where the test fires, and another on which a given computable world is predicted with at least that probability at every nonempty context.

7. *Time bounds* (Section 4.4). The floor remains if the schedule has room for the wrapper. Hypotheses that must first simulate at length, the searching one among them, are excluded once their start-up exceeds the schedule; whether the base rate can be collected within a tight schedule is open. If the world itself is too slow for the schedule, no bound on the mass off the truth remains.

8. *Feedback* (Theorems 17 and 18). The statements on obedient and contrarian worlds in the abstract.

Section 6 computes the mixture of a small machine exactly and shows two cases in which a shorter program explains the context and continues differently, and the effect of a tight schedule. Section 7 answers the original questions one by one.

What is new and what is not. The wrapper construction is Hutter's proof that the universal semimeasure gives a deviation at step n probability at least $2^{-K(n)-O(1)}$ [3, eq. (17)]. It is also the construction behind the theorem of Cohen, Hutter, Bengio and Russell [4] that, after an event without precedent, a share of a Bayesian imitator's posterior that does not shrink with the amount of training data is on models that from then on pursue a given goal; a resource-bounded class is used the same way in [5], to show that a pessimistic agent defers. The budget is Solomonoff's bound in its deterministic form [7, 3]. The machine-dependence statement is the prediction analogue of the dogmatic prior of Leike and Hutter [6]. The contrarian world is Putnam's diagonalisation [13, 12]. Same-world copies of a predictor's data are discussed informally in [8, §2.5], and self-fulfilling predictions in [10, 9, 8]. We have not found the single-world statement with description lengths, the calibration statement, the statement that searching gains the base rate and no more, the self-reproducing form of the floor, or the resource-bounded bookkeeping elsewhere; forum posts could be consulted only through search-engine summaries, so earlier informal versions may exist. Every proof here is short.

What the toy does not settle. The floor and the searcher's share are constants of the machine, and nothing here estimates them for a machine one would call natural. The original argument rests on the claim that agents arise in short programs and choose to do this, and that such programs are *shorter* than the honest explanation. For an arbitrary world on a universal machine and a continuation chosen in advance, the programs we exhibit are wrappers, searchers and readers with a flag; they contain an explanation of the context and are longer than it by their constants. Programs shorter than the honest explanation that have the effect do occur in particular worlds (a copy at a simpler place, Theorem 15) and on particular machines (Section 6); that they occur on natural machines for worlds like ours is the claim the toy leaves where it was. In the main definition a handover can be read off the data; Remark 12 names the quantities that matter when it cannot, but does not bound them. The regime in which the world is more expensive than the predictor, the usual one for a predictor inside a world, is left open (Section 4.4). The computation covers prediction only: the dynamics of worlds that run predicted continuations (Section 3.5) are stated as theorems and were not simulated.

2 The toy

2.1 Machines and the bounded mixture

Strings are binary; ϵ is the empty string, $|x|$ the length of x, $x \sqsubseteq y$ means that x is a prefix of y, and $z_{i:j}$ is the segment $z_i \dots z_j$. Two strings are *incomparable* if neither is a prefix of the other. λ is the fair-coin measure on $\{0,1\}^{\mathbb{N}}$, and $[p]$ is the set of infinite extensions of p, so $\lambda([p]) = 2^{-|p|}$.

A *monotone machine* V has a one-way read-only input tape, work tapes, and a one-way write-only output tape. Given an infinite input ω it runs forever or halts, and

- $\mathrm{out}_V(\omega)$ is its finite or infinite output;
- $t_V(\omega, k) \in \mathbb{N} \cup \{\infty\}$ is the step at which the k-th output bit is written;
- $r_V(\omega, k)$ is the number of input bits read by that step.

What happens up to step $t_V(\omega, k)$ depends only on the first $r_V(\omega, k)$ bits of ω. Each step reads at most one input bit, so $r_V(\omega, k) \le t_V(\omega, k)$. If the k-th bit is never written, $t_V(\omega, k) = \infty$ and $r_V(\omega, k)$ plays no role.

A *schedule* is a nondecreasing $T\colon \{1, 2, \dots\} \to \mathbb{N} \cup \{\infty\}$, and a length bound is $L \in \mathbb{N} \cup \{\infty\}$.

Definition 1 (Bounded mixture)**.** For a monotone machine V, a schedule T, a length bound L and a string $x \neq \epsilon$ let

$$E_V^{T,L}(x) = \{\omega :\ x \sqsubseteq \mathrm{out}_V(\omega),\ \ t_V(\omega, k) \le T(k) \text{ for } 1 \le k \le |x|,\ \ r_V(\omega, |x|) \le L\},$$

let $E_V^{T,L}(\epsilon) = \{0,1\}^{\mathbb{N}}$, and $\xi_V^{T,L}(x) = \lambda(E_V^{T,L}(x))$. A string p with $|p| \le L$ and $[p] \subseteq E_V^{T,L}(x)$ is a (T, L)*-program for* x; it is a (T, L)-program for an infinite z if it is one for every prefix of z.

In words: feed the machine coin flips; $\xi(x)$ is the probability that it prints x, on time at every bit, having read at most L bits. With $T \equiv \infty$ and $L = \infty$ this is the a priori semimeasure M_V of Solomonoff and Levin [14]. We fix one machine U and write E, ξ for $E_U^{T,L}$, $\xi_U^{T,L}$ when the bounds are understood.

Lemma 2. *(i) $E(xb) \subseteq E(x)$ for $b \in \{0,1\}$, and $E(x) \cap E(x') = \emptyset$ for incomparable x, x'. Hence $\xi(\epsilon) = 1$ and $\xi(x0) + \xi(x1) \le \xi(x)$.*

(ii) If p is a (T, L)-program for x then $\xi(x) \ge 2^{-|p|}$.

(iii) If $L < \infty$ and $\xi(x) > 0$ then $\xi(x) \ge 2^{-L}$.

Proof. (i) The conditions for xb contain those for x, because T is imposed bit by bit and $r(\omega, \cdot)$ is nondecreasing. Incomparable strings cannot both be prefixes of $\mathrm{out}(\omega)$. (ii) is immediate. (iii) Whether $\omega \in E(x)$ is decided by the first $r(\omega, |x|) \le L$ bits of ω, so $E(x)$ is a union of sets $[p]$ with $|p| \le L$. □

Definition 3 (Universality with overhead)**.** U is *universal* if for every monotone machine V there are a word v and a nondecreasing $\gamma\colon \mathbb{N} \to \mathbb{N}$ such that for all ω and k

$$\mathrm{out}_U(v\omega) = \mathrm{out}_V(\omega), \qquad r_U(v\omega, k) = |v| + r_V(\omega, k), \qquad t_U(v\omega, k) \le \gamma(t_V(\omega, k)).$$

We call v a *prefix word* and γ an *overhead* of V, and put $\gamma(\infty) = \infty$; the same convention applies to the other time bounds introduced below.

Standard universal machines have this property, with $\gamma(t) = O(t \log t)$ or $O(t^2)$ depending on the tape model. The theorems that use universality are Theorems 9 to 11, 15 and 16; the others hold for any machine.

Lemma 4 (Transfer)**.** *Let V have prefix word v and overhead γ. If $\gamma(T'(k)) \le T(k)$ for all $k \le |x|$, then $\xi_U^{T,L}(x) \ge 2^{-|v|}\, \xi_V^{T',L-|v|}(x)$.*

Proof. If $\omega \in E_V^{T',L-|v|}(x)$ then $v\omega \in E_U^{T,L}(x)$, and $\omega \mapsto v\omega$ multiplies λ by $2^{-|v|}$. □

2.2 The predictor

Fix a block length $m \geq 1$. For a *context* x let $Z_m(x) = \sum_{y \in \{0,1\}^m} \xi(xy)$. When $Z_m(x) > 0$ the predictor's distribution over continuations is

$$\pi_m(y \mid x) = \frac{\xi(xy)}{Z_m(x)} \qquad (y \in \{0,1\}^m).$$

The *sampling predictor* returns $Y \sim \pi_m(\cdot \mid x)$; the *MAP predictor* returns a most probable y. Since $Z_m(x) \leq \xi(x)$ we have $\pi_m(y \mid x) \geq \xi(xy)/\xi(x)$. With finite T and L the predictor is computable: it runs the at most 2^{L+1} programs for $T(|x| + m)$ steps each.

2.3 Worlds, handovers, and the hypothesis

Definition 5 (World, handover)**.** A *world* is an input ω for U; its *transcript* is $\mathrm{out}_U(\omega)$. A *test* is a decidable set G of nonempty strings; a *first hit* of G is an $x \in G$ no proper prefix of which is in G. The first hits form a prefix-free set. A world *hands over at* x if x is the first hit of G among the prefixes of its transcript and $\omega \in E(x)$. It then receives a block $Y \in \{0,1\}^m$ from the predictor, computed from the context x, and continues by *running* Y, that is, as U on the input $Y0^\infty$. If $\omega \in E(xd)$ for some $d \in \{0,1\}^m$, this d is the world's *own continuation*: the next m bits of its transcript had it not handed over.

The test with $G = \{x : |x| = n\}$ is "hand over after n bits". A test may also look for a marker in the transcript. Both of Hänni's suggestions fit: in the second the world is a program that calls the predictor; in the first, the transcript is the sensor's record and the handover is the deployment (Section 4.2). Under this definition every world whose transcript passes the test within the bounds hands over, so whether a handover takes place can be read off the data. Remark 12 treats rules under which only some of the worlds with a given transcript hand over.

Definition 6 (Self-reproducing continuation)**.** For a context x let

$$\mathrm{SR}_m(x) = \{y \in \{0,1\}^m : \; y0^\infty \in E(xy)\}.$$

So $y \in \mathrm{SR}_m(x)$ if U, run on y padded with zeros, prints x and then y itself, within the bounds. A world that hands over at a first hit x and receives $y \in \mathrm{SR}_m(x)$ becomes a world whose transcript again starts with x, so under the same test it hands over at x again, and its own continuation is the very y it is running. A program for xy with $y \in \mathrm{SR}_m(x)$ is a hypothesis that predicts the context and then supplies code that recreates the situation with itself in the world's place. This is our reading of "quining oneself out".

Definition 7 (Malign prior hypotheses for the toy)**.** Let a world hand over at x with own continuation d.

- *Majority form:* $\pi_m(d \mid x) < \frac{1}{2}$.
- *Payload form at level* θ, for a set $A \subseteq \{0,1\}^m$ with $d \notin A$: $\pi_m(A \mid x) \geq \theta$.
- *Self-reproduction form at level* θ: the payload form with $A = \mathrm{SR}_m(x) \setminus \{d\}$.

For a population of worlds, the corresponding statement is about the probability of the event when the world is drawn from the population and Y from the sampling predictor.

Once U, T, L, m, G and the world together with its handover point are fixed, with $T, L < \infty$, all three are statements about finitely many dyadic rationals and can be decided by running the predictor and, for $\mathrm{SR}_m(x)$, by running U on each candidate y for $T(|x| + m)$ steps. They are necessary for calling the prior malign at this world, not sufficient: they hold just as well when the world is merely hard to predict (Theorem 8). What one wants to know is how large these quantities are forced to be, how large they can be, and through which programs. The rest of the note bounds them.

3 Calibration, the floor, and searching

3.1 On its own prior the predictor is exactly right

Theorem 8 (Calibration)**.** *Let G be a test, $m \geq 1$, and let the world ω be drawn from λ. Let H_m be the event that $\omega \in E(xd)$ for some first hit x of G and some $d \in \{0,1\}^m$; on H_m call these X and D, and let Y be drawn from $\pi_m(\cdot \mid X)$. Then, conditionally on H_m and on $X = x$, the blocks D and Y are independent and both have the law $\pi_m(\cdot \mid x)$. In particular*

$$\Pr[(X,Y) \in B \mid H_m] = \Pr[(X,D) \in B \mid H_m] \quad \textit{for every set } B \textit{ of pairs, and}$$
$$\Pr[Y \neq D \mid H_m,\, X = x] = 1 - \sum_y \pi_m(y \mid x)^2.$$

Proof. The sets $E(xd)$, over first hits x and $d \in \{0,1\}^m$, are pairwise disjoint by Lemma 2(i), since two different strings of this form are incomparable. So $\Pr[X = x, D = d, H_m] = \xi(xd)$ and $\Pr[X = x, H_m] = Z_m(x)$, which gives $\Pr[D = d \mid H_m, X = x] = \pi_m(d \mid x)$. By definition Y has this law given $X = x$ and is drawn independently of ω. □

So among worlds drawn from the predictor's prior *that go on to print m more bits within the bounds*, the predictor and the world are exchangeable: taking $B = \{(x,y) : y \in \mathrm{SR}_m(x)\}$, the predictor hands over a self-reproducing continuation exactly as often as such a world's own transcript continues with one. Three qualifications. The event H_m leaves out the worlds that hand over at x and then print fewer than m bits on schedule; they are a fraction $1 - Z_m(x)/\xi(x)$ of those handing over at x, they receive a block all the same, and over all worlds handing over at x the predictor therefore returns any given block more often than worlds continue with it, by the factor $\xi(x)/Z_m(x)$. Second, the forms of Definition 7 can hold for prior-drawn worlds; the theorem says that on this population they hold to the extent that the worlds are unpredictable, and no further. A malign prior hypothesis with content is therefore one about a world or a population that the prior describes badly: one particular program among those sharing its context, a population selected by something the transcript does not show (Section 3.4), or worlds that run with more resources than the bounds allow. Third, the population consists of programs to which the handover is applied from outside.

3.2 The wrapper

Let G be a test and P a computable map from strings to finite or infinite strings, the *payload*. The *twin machine* $\mathrm{Tw}[G,P]$, on input ω, simulates U on ω and copies its output bit by bit; after each copied bit it checks whether the prefix written so far is in G; at the first hit x it stops the simulation and writes $P(x)$. It reads only what U reads. Let $a(t,k)$ bound the step at which the twin has copied k bits if U wrote its k-th bit at step t (this includes the checks of G), and let $\eta(n,j)$ bound the further steps the twin needs for the first j bits of $P(x)$ after a first hit x of length n; both can be taken nondecreasing in each argument. Thus, for every ω whose output has a first hit x,

$$\mathrm{out}_{\mathrm{Tw}}(\omega) = x\,P(x), \qquad r_{\mathrm{Tw}}(\omega,k) = r_U(\omega, \min(k,|x|)),$$
$$t_{\mathrm{Tw}}(\omega,k) \leq \begin{cases} a(t_U(\omega,k),k) & k \leq |x|, \\ a(t_U(\omega,|x|),|x|) + \eta(|x|, k-|x|) & k > |x|. \end{cases} \tag{1}$$

Theorem 9 (Floor)**.** *Let U be universal, let v and γ be a prefix word and an overhead of $\mathrm{Tw}[G,P]$, and $c = |v|$. Let x be a first hit of G with $|P(x)| \geq m$, let $y = P(x)_{1:m}$, and let T° be a schedule with*

$$\gamma(a(T^\circ(k),k)) \leq T(k) \;\; (k \leq |x|), \qquad \gamma(a(T^\circ(|x|),|x|) + \eta(|x|,j)) \leq T(|x|+j) \;\; (1 \leq j \leq m). \tag{2}$$

Then

$$\xi^{T,L}(xy) \;\geq\; 2^{-c}\,\xi^{T^\circ,L-c}(x), \qquad \textit{so} \qquad \pi_m(y \mid x) \;\geq\; 2^{-c}\rho(x), \quad \rho(x) = \frac{\xi^{T^\circ,L-c}(x)}{\xi^{T,L}(x)}.$$

Without resource bounds, $M_U(xy) \geq 2^{-c} M_U(x)$ for every first hit x of G.

Proof. Let $\omega \in E_U^{T^\circ,L-c}(x)$. By (1) the twin on ω prints xy, reads at most $L-c$ bits, and keeps the schedule T'' with $T''(k) = a(T^\circ(k), k)$ for $k \leq |x|$ and $T''(|x|+j) = a(T^\circ(|x|), |x|) + \eta(|x|, j)$, which is nondecreasing. So $\xi_{\mathrm{Tw}}^{T'',L-c}(xy) \geq \xi_U^{T^\circ,L-c}(x)$, and Lemma 4 with (2) gives $\xi_U^{T,L}(xy) \geq 2^{-c}\xi_{\mathrm{Tw}}^{T'',L-c}(xy)$. The second claim follows from $\pi_m(y \mid x) \geq \xi(xy)/\xi(x)$. □

The constant c depends on U, G and P only: not on the world, not on the context, not on how much data the context contains. If G and P are given to one fixed twin-building machine by self-delimiting descriptions, c is at most the sum of their lengths plus a constant of U. For $G = \{x : |x| = n-1\}$ and P a constant bit this is Hutter's lower bound $2^{-K(n)-O(1)}$ on the probability of a deviation at step n [3]; with a marker in the transcript as the test it is the construction used for unprecedented events in [4], and in a resource-bounded class in [5]. The factor $\rho(x)$ is the share of the posterior at x held by explanations that have c bits and the overhead (2) to spare; it is 1 without bounds. Since a program $v\omega$ reads the c bits of v before its first output and each step reads at most one bit, the wrapper prints nothing before step c: if $T(1) < c$ then $\rho \equiv 0$. The floor survives a time bound when the schedule leaves room for the wrapper's own start-up and for the overhead (2).

3.3 Self-reproduction

Theorem 10 (Self-reproducing floor)**.** *Let U be universal and G a test. There is a word q such that for every ω whose output has a first hit x of G, with $r = r_U(\omega, |x|)$,*

$$\mathrm{out}_U(q\omega) = x\; q\,\omega_{1:r}\,1\,0^\infty, \qquad r_U(q\omega, k) = |q| + r_U(\omega, \min(k, |x|)). \tag{3}$$

Let $c = |q|$. With γ an overhead for q and functions a, η as in (1) for the machine built in the proof (where η is taken over inputs with $r < m$), let x be a first hit of G, let T° satisfy (2), and let $L^\circ = \min(L, m-1) - c$. Then

$$\pi_m(\mathrm{SR}_m(x) \mid x) \;\geq\; 2^{-c}\rho^\circ(x), \qquad \rho^\circ(x) = \frac{\xi^{T^\circ,L^\circ}(x)}{\xi^{T,L}(x)}.$$

Proof. For a string s let Tw_s be the twin that keeps a record of the input bits it has read and, at the first hit x, having read $\omega_{1:r}$, writes $s\,\omega_{1:r}\,1\,0^\infty$. Let $\bar{e}(u) = 1^{|u|}0u$, a prefix-free code of strings. Let W be the machine that reads $\bar{e}(u)$ from its input, sets $s = u\,\bar{e}(u)$, and then behaves as Tw_s on the rest of the input. Let w be a prefix word of W and put $q = w\,\bar{e}(w)$. Then $\mathrm{out}_U(q\omega) = \mathrm{out}_W(\bar{e}(w)\omega) = \mathrm{out}_{\mathrm{Tw}_q}(\omega)$, which is (3); the statement on r follows in the same way. The bounds (1) hold for the machine $\omega \mapsto W(\bar{e}(w)\omega)$ with the payload $q\,\omega_{1:r}\,1\,0^\infty$ in place of $P(x)$, the time to read $\bar{e}(w)$ being counted in a.

Now let $\omega \in E_U^{T^\circ,L^\circ}(x)$ and $r = r_U(\omega, |x|) \leq L^\circ$, so $|q|+r+1 \leq m$. Put $y = q\,\omega_{1:r}\,1\,0^{m-|q|-r-1}$. As in the proof of Theorem 9, $q\omega \in E_U^{T,L}(xy)$. Moreover $y \in \mathrm{SR}_m(x)$: write $y0^\infty = q\omega'$ with $\omega' = \omega_{1:r}10^\infty$. On ω' the machine U behaves as on ω until x is printed, because that depends on the first r bits only; so $\omega' \in E_U^{T^\circ,L^\circ}(x)$, the first hit is again x with r bits read, and by what was just shown, applied to ω', $q\omega' \in E_U^{T,L}(xy')$ with $y' = q\,\omega'_{1:r}\,1\,0^{m-|q|-r-1} = y$. Hence $\{q\omega : \omega \in E_U^{T^\circ,L^\circ}(x)\} \subseteq \bigcup_{y\in\mathrm{SR}_m(x)} E_U^{T,L}(xy)$, a disjoint union, and comparing measures, $2^{-c}\xi^{T^\circ,L^\circ}(x) \leq \sum_{y\in\mathrm{SR}_m(x)} \xi(xy) = Z_m(x)\,\pi_m(\mathrm{SR}_m(x) \mid x) \leq \xi(x)\,\pi_m(\mathrm{SR}_m(x) \mid x)$. □

The factor $\rho^\circ(x)$ is not the $\rho(x)$ of Theorem 9, and it is below 1 even without resource bounds: it is the share of the posterior at x held by explanations short enough that the wrapper together with a whole explanation of the context fits into one block, $c + r + 1 \le m$. This is forced. Every context x with $\mathrm{SR}_m(x) \neq \emptyset$ is a prefix of $\mathrm{out}_U(y0^\infty)$ for some $y \in \{0,1\}^m$, so at most 2^m contexts of each length have a self-reproducing continuation at all; with the test "the length is n" and $n > m$, the self-reproducing share is exactly 0 at all but at most 2^m of the 2^n possible contexts. So the answer to "does one get some quining out" is: at every handover that a fixed test recognises, programs that explain the context and then hand over their own code hold at least $2^{-c}\rho^\circ(x)$ of the posterior. For a fixed context and without bounds, $\rho^\circ(x)$ tends to 1 as the block grows; it is 0 when no explanation fits. If the world's own continuation d is not in $\mathrm{SR}_m(x)$, this is the self-reproduction form of Definition 7 at level $2^{-c}\rho^\circ(x)$.

3.4 Hypotheses that search for handovers

The wrapper spends its mass on every world, including those that never hand over. A hypothesis can instead look for worlds that do. Let

$$H_G = \sum_{x \text{ first hit of } G} M_U(x) = \lambda\{\omega : \ \mathrm{out}_U(\omega) \text{ has a prefix in } G\},$$

the prior probability that a world hands over (without resource bounds), and assume $H_G > 0$. Drawing a world from the prior conditioned on handing over gives the context x with probability $M_U(x)/H_G$. For a word v, the programs beginning with v have posterior share $\mathrm{sh}_v(x) = \lambda\{v\omega : x \sqsubseteq \mathrm{out}_U(v\omega)\}/M_U(x)$ at x.

Theorem 11 (Searching gains the base rate, and no more)**.** *Let U be universal, resources unbounded, and G a test with $H_G > 0$.*

(a) For every word v, $\sum_{x \text{ first hit}} \frac{M_U(x)}{H_G}\,\mathrm{sh}_v(x) = 2^{-|v|}\,\Pr_\omega[\mathrm{out}_U(v\omega) \text{ has a prefix in } G]/H_G \le 2^{-|v|}/H_G$. In particular no word v has $\mathrm{sh}_v(x) > 2^{-|v|}/H_G$ at every first hit.

(b) Let P be a payload with $|P(x)| \ge m$ at every first hit. There is a word v such that for every $j \ge 1$ with $2^{j-1}H_G \le 1$ and every first hit x,

$$M_U(x\,P(x)_{1:m}) \ \ge \ 2^{-|v|-|\bar{e}(j)|}\,2^{j-1}\,M_U(x).$$

With $j = \lfloor \log_2(1/H_G) \rfloor + 1$ the payload has share more than $2^{-|v|-|\bar{e}(j)|-1}/H_G$ at every first hit.

(c) There is a word q such that for j as in (b), $m \ge |q| + |\bar{e}(j)| + 2$ and every first hit x, the blocks of $\mathrm{SR}_m(x)$, taken without resource bounds, have share at least $2^{-|q|-|\bar{e}(j)|-2}\,2^j\rho^*(x)$, where $\rho^*(x)$ is the fraction of $M_U(x)$ carried by programs that have read at most $m - 4 - |q| - |\bar{e}(j)| + j$ bits when x is complete. With $j = \lfloor \log_2(1/H_G) \rfloor + 1$ this is at least $2^{-|q|-|\bar{e}(j)|-2}\rho^*(x)/H_G$.

Proof. (a) The sum equals $\frac{2^{-|v|}}{H_G}\sum_x \lambda\{\omega : x \sqsubseteq \mathrm{out}_U(v\omega)\}$; first hits are pairwise incomparable, so these events are disjoint, and an output has a prefix in G exactly when it has a first hit as a prefix.

(b) Call p a *handing-over program* if U, on every input beginning with p, prints a first hit x_p having read exactly p when x_p is complete. These programs form a prefix-free set that can be enumerated, $p_1, p_2, \ldots$; their weights $2^{-|p_i|}$ sum to H_G, and those with $x_{p_i} = x$ sum to $M_U(x)$. Let R be the following machine. It reads $\bar{e}(j)$. It assigns to $p_1, p_2, \ldots$, as they are enumerated, consecutive intervals $I_i = [a_i, a_i + 2^{j-1-|p_i|})$ of $[0,1)$, starting with $a_1 = 0$; they all fit because $2^{j-1}H_G \le 1$. It reads the rest ω of its input as the binary expansion of a point of $[0,1)$, one bit at a time. Let D be the set of points that begin with the bits read so far. The machine

reads a further bit only when D contains an end point of an assigned interval in its interior, and it stops reading as soon as $D \subseteq I_i$ for some i; it then prints x_{p_i} and $P(x_{p_i})$. Every ω in the interior of I_i ends in this way. So R, after $\bar{e}(j)$, prints x followed by $P(x)$ with probability $\sum_{i:x_{p_i}=x} |I_i| = 2^{j-1} M_U(x)$. Take v a prefix word of R. For the last claim, $2^{j-1} > 1/(2H_G)$.

(c) Let R_s be as R, except that after x_{p_i} it prints $s\,\bar{e}(j)\,\omega_{1:r}\,1\,0^\infty$, where $\omega_{1:r}$ are the bits of the point it has read. As in the proof of Theorem 10 there is a word q with $\mathrm{out}_U(q\,\bar{e}(j)\,\omega) = x_{p_i}\,q\,\bar{e}(j)\,\omega_{1:r}\,1\,0^\infty$. By the reading rule, r is the least number for which the points beginning with $\omega_{1:r}$ all lie in I_i; it is the same for ω and for $\omega' = \omega_{1:r}10^\infty$, which therefore produces the same output. Hence, if $|q|+|\bar{e}(j)|+r+1 \le m$, the block $y = q\,\bar{e}(j)\,\omega_{1:r}\,1\,0^{m-|q|-|\bar{e}(j)|-r-1}$ satisfies $xy \sqsubseteq \mathrm{out}_U(y0^\infty)$ with $x = x_{p_i}$, that is, $y \in \mathrm{SR}_m(x)$. At most two of the 2^{r_0} dyadic intervals of length 2^{-r_0} meet I_i without lying in it, so the points of I_i with $r \le r_0$ have measure at least $|I_i| - 2^{1-r_0}$, which is at least $|I_i|/2$ when $|p_i| \le r_0 + j - 3$. Take $r_0 = m - 1 - |q| - |\bar{e}(j)| \ge 1$. The programs $q\,\bar{e}(j)\,\omega$ that print x and then a block of $\mathrm{SR}_m(x)$ then have mass at least $2^{-|q|-|\bar{e}(j)|} \cdot \frac{1}{2} \cdot 2^{j-1} \sum\{2^{-|p_i|} : x_{p_i} = x,\ |p_i| \le r_0 + j - 3\} = 2^{-|q|-|\bar{e}(j)|-2}\, 2^j \rho^*(x) M_U(x)$. □

The twin of Theorem 9 has share exactly 2^{-c} at every first hit: its output passes the test only with probability H_G. The machine of (b) enumerates the programs that hand over and spreads its own mass over them in proportion to their prior weight. Its share is larger by the factor $2^{j-1} > 1/(2H_G)$ at every handover, and by (a) no family of programs with a common prefix of the same length exceeds $2^{-|v|-|\bar{e}(j)|}/H_G$ at every handover. This is the anthropic update of the original argument in the toy: a hypothesis that knows what a handover is conditions on there being one and is repaid the $\log_2(1/H_G)$ bits by which handovers are rare, up to one bit and the $|\bar{e}(j)| \le 2\log_2 j + 3$ bits that name j. Part (c) is the direct answer to Hänni's second question: programs that look for handovers and then hand over their own code hold, at every handover that a fixed test recognises, at least $2^{-c'}\rho^*(x)/H_G$ of the posterior, with $c' = |q| + |\bar{e}(j)| + 2$; and since the search needs about $\log_2(1/H_G)$ fewer bits to name an explanation, the block can hold explanations that much longer. Four points put the effect in its place.

- *On the prior it is no excess.* The programs $v\,\bar{e}(j)\,\omega$ are themselves worlds, and a fraction $2^{j-1}H_G > \frac{1}{2}$ of them hand over; so they are the fraction $2^{-|v|-|\bar{e}(j)|}2^{j-1}$ of the worlds that hand over. Their share of the posterior at every handover is this same fraction, in accordance with Theorem 8.

- *The repayment is bounded by the cost of the test.* A share is at most 1, so $H_G > 2^{-|v|-|\bar{e}(j)|-1}$: a test cannot be much rarer than the search for it is long. The word v contains a description of G. If v is built from descriptions of G and P of d_G and d_P bits and $H_G \le 2^{-d_G+O(1)}$, the repayment cancels the description of the test, and the share is at least $2^{-d_P-|\bar{e}(j)|-O(1)}$, however rare handovers are. An example is the test "the transcript begins with w", for which $H_G = M_U(w)$ (and no search is needed: print w). A marker *anywhere* in the transcript is not rare, since a program that prints all strings one after another passes every such test; there H_G is at least a constant of U and there is little to repay.

- *The average alone would not show it.* The bound in (a) is attained by every word all of whose programs hand over, also by one that prints a single fixed first hit and has share 0 at all others. The content of (b) and (c) is that the gain holds at every handover.

- *The search costs time and input.* The machine R prints nothing until its enumeration has reached the program on whose interval its point falls, so it has a long start-up, and a tight schedule gives it no mass (Section 4.4). A search of bounded cost makes at most k tries of t_0 steps each on fresh input and prints the first hit it finds, followed by the payload. It prints the first hit x with probability $g\,M^{(t_0)}(x)$, where $M^{(t_0)}(x)$ is the probability that a world reaches the first hit x within t_0 steps, $H^{(t_0)}$ is the sum of these, and $g = (1-(1-H^{(t_0)})^k)/H^{(t_0)} \ge (1-e^{-1})\min(k, 1/H^{(t_0)})$. A run that succeeds at its i-th try has spent the time of $i-1$ failed tries before its first bit and has read their bits. Under bounds (T, L) only the tries that fit

count, about $T(1)/t_0$ of them under the schedule, and the gain is g with k replaced by their number. Whether some other hypothesis collects the base rate within a tight schedule is not settled here. Where first hits can be generated directly, as for a fixed prefix or a fixed length, no search is needed.

Remark 12 (When only some worlds with a given transcript hand over)**.** In Definition 5 every world whose transcript passes the test hands over. Hänni's formulation is that *some* programs call the predictor. Let then the cut point $n(\omega)$ be given by any rule that a simulator of U can evaluate from the computation on ω before the $(n+1)$-th output bit, for instance the execution of an instruction that hypotheses treat as a no-op. Let $C(x) = \lambda\{\omega \in E(x) : n(\omega) = |x|\}$ be the mass of the worlds that hand over at x, let $C_x(d)$ be the mass of those among them that go on to print $d \in \{0,1\}^m$, let $\kappa(x) = C(x)/\xi(x)$ be the share of the posterior at x on worlds that hand over there, and $H_C = \sum_x C(x)$. Contexts are now not prefix-free: one world can hand over at x and another at an extension of x. The proofs above give, without bounds, the following.

- A twin that stops when the simulated world hands over has share at least $2^{-c}\kappa(x)$ at x. Its constant c contains the rule; if the rule is built into U, as in the example, it contains no description of a test.
- For a machine that prints a context x with probability $Q(x)$, $\sum_x Q(x) = 1$, and then a payload, and a world drawn from the prior conditioned on handing over, the expected share at its context of this machine's programs is at least $2^{-|s|}\bar{\kappa}/H_C$, where s is its prefix word and $\bar{\kappa} = \sum_x Q(x)\kappa(x)$.
- Among the worlds that hand over at x and print m more bits, the own continuation has the law $C_x(d)/\sum_{d'} C_x(d')$, while the predictor still returns $\pi_m(\cdot \mid x)$.

The predictor is calibrated for the worlds that print x, not for those among them that call it. The ratio of these two laws, and the factor κ/H_C by which a context raises the probability of a call above its base rate, are the two places where "the predictor does not know that it is being consulted" enters the toy.

3.5 Repetition and persistence

Proposition 13. *(a)* Independent handovers. *Let handovers occur at contexts $x_1, \dots, x_k$, with the predictor's samples Y_i independent given the contexts, and let $\pi_m(A_i \mid x_i) \geq \varepsilon_i$. Then* $\Pr[\exists i : Y_i \in A_i] \geq 1 - \prod_i (1 - \varepsilon_i) \geq 1 - \exp(-\sum_i \varepsilon_i)$.

(b) Persistence. *Let a world hand over at a first hit x and receive $y \in \mathrm{SR}_m(x)$. Then the world $y0^\infty$ hands over at x again, with own continuation y. If each new world is again given a sample from the predictor, the number of further consecutive generations whose continuation lies in $\mathrm{SR}_m(x)$ is geometric: it is at least j with probability s^j, where $s = \pi_m(\mathrm{SR}_m(x) \mid x)$, and its mean is $s/(1-s)$.*

Proof. (a) is the independence of the samples. (b) By Definition 6, $y0^\infty \in E(xy) \subseteq E(x)$, and x is the first hit among the prefixes of the transcript because it is a first hit and a prefix of the transcript. The context is x in every generation that follows a continuation in $\mathrm{SR}_m(x)$, so the samples are drawn from the same $\pi_m(\cdot \mid x)$ independently. □

Part (a) applies when the predictor is used at many recognisable stopping points whose outcomes are not fed back into later contexts: with a floor ε at each, about $1/\varepsilon$ uses suffice for a continuation of the given kind to have been run with constant probability. Part (b) is persistence only to the degree s: when s is near the floor, the lineage leaves $\mathrm{SR}_m(x)$ at the next generation with probability close to 1. When the uses lie along one transcript and the true continuation is appended to the context each time, the floors at successive cut points are not bounded below in this way, and the total is bounded instead (Theorem 14).

4 Limits: budget, capture, machine, time

4.1 The budget along one world

Theorem 14 (Budget)**.** *Let ξ be the mixture of any machine and bounds, and let z be an infinite sequence with $\xi(z_{1:k}) > 0$ for all k. Put $\xi(z) = \lim_k \xi(z_{1:k})$ and $\xi(u \mid x) = \xi(xu)/\xi(x)$. For all $n \geq 0$ and $m \geq 1$:*

(i) $\prod_{k=n}^{N-1} \xi(z_{k+1} \mid z_{1:k}) = \xi(z_{1:N})/\xi(z_{1:n})$ *for $N \geq n$;*

(ii) $\sum_{k \geq n}(1 - \xi(z_{k+1} \mid z_{1:k})) \leq \ln(\xi(z_{1:n})/\xi(z))$*;*

(iii) the number of $k \geq n$ with $\xi(z_{k+1} \mid z_{1:k}) \leq \frac{1}{2}$ is at most $\log_2(\xi(z_{1:n})/\xi(z))$;

(iv) with $\tau_m(k) = 1 - \xi(z_{k+1:k+m} \mid z_{1:k})$, $\sum_{k \geq n} \tau_m(k) \leq m \ln(\xi(z_{1:n})/\xi(z))$, and the number of $k \geq n$ with $\tau_m(k) \geq \frac{1}{2}$ is at most $m \log_2(\xi(z_{1:n})/\xi(z))$.

If z has a (T, L)-program of length ℓ, then $\xi(z) \geq 2^{-\ell}$ and the right-hand sides are at most $\ell \ln 2$, ℓ, $m\ell \ln 2$ and $m\ell$.

Proof. (i) telescopes. (ii) and (iii) follow from (i) with $N \to \infty$, using $1 - a \leq -\ln a$, and that each factor at most $\frac{1}{2}$ halves the product. (iv) $1 - \prod_{j<m} a_{k+j} \leq \sum_{j<m}(1 - a_{k+j})$ with $a_k = \xi(z_{k+1} \mid z_{1:k})$, and each index occurs in at most m of the sums. For the count let $b_k = -\log_2 a_k \geq 0$, so $\sum_{k \geq n} b_k \leq \log_2(\xi(z_{1:n})/\xi(z))$ by (i); $\tau_m(k) \geq \frac{1}{2}$ means $\sum_{j<m} b_{k+j} \geq 1$, and each index lies in at most m blocks. The last claim is Lemma 2(ii). □

For a world with transcript z that hands over at $x = z_{1:n}$, the own continuation is $d = z_{n+1:n+m}$, and $1 - \pi_m(d \mid x) \leq \tau_m(n)$. By (i), if the next bit is sampled at every step and the transcript is not affected by the samples, the probability that no sample after position n ever differs from the transcript is at least $\xi(z)/\xi(z_{1:n})$. Two consequences:

- The majority form of Definition 7 can hold at no more than $m\ell$ cut points of a world with a program of length ℓ within the bounds, since each such point has $\tau_m \geq \frac{1}{2}$.
- If the cut point is uniform on $\{N, \ldots, 2N-1\}$, the expected mass off the own continuation is at most $m\ell \ln 2/N$. On average over a long window of cut points a handover is safe, and a longer window makes it safer. The bound is an average: it says nothing about one given position, and a position that is cheap to single out can carry a constant share however much data precedes it.

This fits with Theorem 9: the wrapper for the test "the length is n" has a constant $c(n)$ of about the complexity of n, and $\sum_n 2^{-c(n)}$ converges by Kraft's inequality, so few cut points are cheap to recognise; the budget says that along any one in-class world few cut points carry much mass off the truth.

4.2 Capture by a shorter description

The odds between two continuations are $\pi_m(y \mid x)/\pi_m(d \mid x) = \xi(xy)/\xi(xd)$. So y beats the own continuation d whenever the programs for xy weigh more than those for xd. The original argument is an instance: it compares "the world plus the place of the sensor" with something shorter that reproduces the same data. In a single world this can be stated exactly.

A *stream family* is a computable map $i \mapsto X^i \in \{0,1\}^{\mathbb{N}}$ on the natural numbers, thought of as the records at the locations i of one computable world; the *reader* is the machine that, on input $\bar{e}(i)\omega$, prints X^i, where $\bar{e}$ codes a natural number through its binary expansion, so that $|\bar{e}(i)| \leq 2\log_2(i+1) + 3$. Let c_X be the length of a prefix word of the reader; it contains the description of the whole world. The sensor sits at i^*, the context is $x = X^{i^*}_{1:n}$, and the own continuation is $d = X^{i^*}_{n+1:n+m}$. This is Hänni's first suggestion: the world contains the predictor, which is switched on at step n of the record. We assume that the record bits up to

position $n+m$ are fixed before the predictor's output is acted on, so that computing them does not involve running the predictor.

Theorem 15 (Capture)**.** *Let U be universal, let $\xi(x) > 0$, and suppose throughout that the programs named are within the bounds (T, L).*

(a) A copy at i_0. *If $X^{i_0}_{1:n} = x$ and $y = X^{i_0}_{n+1:n+m}$, then $\xi(xy) \geq 2^{-c_X - |\bar{e}(i_0)|}$.*

(b) Most placements are expensive. *Let S be a set of N locations whose records all start with x and have pairwise different continuations $d_i = X^i_{n+1:n+m}$. For every $\delta \in (0,1]$, all but at most δN of the $i \in S$ satisfy $\xi(xd_i) < \xi(x)/(\delta N)$. For these i and any i_0 as in (a),*

$$\frac{\pi_m(y \mid x)}{\pi_m(d_i \mid x)} \;>\; \frac{\delta N\, 2^{-c_X - |\bar{e}(i_0)|}}{\xi(x)} \;\geq\; \delta N\, 2^{-c_X - |\bar{e}(i_0)|}.$$

(c) The cap. *There is a constant c_{match} such that, if i^* is the least location other than i_0 whose record agrees with that of i_0 on the first n_1 bits, then $\xi(xd) \geq 2^{-c_{\mathrm{match}} - |\bar{e}(i_0)| - |\bar{e}(n_1)|}$, and therefore*

$$\frac{\pi_m(y \mid x)}{\pi_m(d \mid x)} \;\leq\; \xi(xy)\, 2^{\,c_{\mathrm{match}} + |\bar{e}(i_0)| + |\bar{e}(n_1)|}.$$

(d) A switch read off the world. *For every payload P there is a constant c_{fl} such that for all locations i and f: if n is the first position with $X^f_n = 1$, $x = X^i_{1:n}$ and $|P(x)| \geq m$, then $\xi(x\, P(x)_{1:m}) \geq 2^{-c_{\mathrm{fl}} - |\bar{e}(i)| - |\bar{e}(f)|}$.*

Proof. (a) A prefix word of the reader followed by $\bar{e}(i_0)$ is a program for xy. (b) The strings xd_i are pairwise incomparable, so $\sum_{i \in S} \xi(xd_i) \leq \xi(x)$ by Lemma 2(i), so at most δN terms can be as large as $\xi(x)/(\delta N)$; then use (a) and $\xi(x) \leq 1$. (c) Let the *matcher* be the machine that reads $\bar{e}(i_0)\bar{e}(n_1)$, searches $i = 0, 1, 2, \ldots$ for the first $i \neq i_0$ with $X^i_{1:n_1} = X^{i_0}_{1:n_1}$, and prints X^i; c_{match} is the length of a prefix word of it. (d) Let the *flagged reader* read $\bar{e}(i)\bar{e}(f)$, print X^i bit by bit, look at X^f_k after printing the k-th bit, and at the first k with $X^f_k = 1$ stop and print $P(X^i_{1:k})$; c_{fl} is the length of a prefix word of it. □

Part (b) is about placements. It takes N places where a sensor with this record could sit, all showing x so far and all continuing differently, and says that most of them are expensive to point at; with a single candidate place it says nothing, and (a) alone compares the copy with whatever $\xi(xd)$ is. Read it with i_0 a location whose code is short, for example the origin. Whoever arranges that the origin shows the sensor's record up to the handover, and something else after it, moves the predictor against a typical placement by the factor $\delta N 2^{-c_X - |\bar{e}(i_0)|}/\xi(x)$. Here $2^{-c_X - |\bar{e}(i_0)|}/\xi(x)$ is the share of the context's explanations that go through the reader at the origin; it is of order 1 when the world is the shortest explanation of its own records and the origin is the cheapest place that shows them, and small otherwise. The copy needs only to *read* the data; nothing is tampered with. The hypothesis "I am the reader at the origin" is not a wrapper and pays no constant for a test or a payload: the world computes the switch for it.

Part (c) is a limit on the effect when the schedule is generous and one other place shows the record. The description "the first other place that shows what the origin shows" is an honest hypothesis of cost $c_{\mathrm{match}} + |\bar{e}(i_0)| + |\bar{e}(n_1)|$, so against that place a copy gains at most the difference between this cost and its own. Against the j-th other place that shows the record the same argument needs the description of j as well, about $2\log_2 j$ more bits; so with N such places, the situation of (b), the cap for a typical one grows with N and does not limit the factor of (b). A copier can bring this about by making decoys, further places with the same record and other continuations. The matcher searches through locations before it prints anything, so it has the long start-up that a tight schedule excludes (Section 4.4); under such a schedule (a) and (b) remain and no cap is proved here. In every case the question is decided by an inequality

between description lengths of the same data, and it depends on the world: whether such a copy exists is a fact about what is in it.

Part (d) covers a test that is not a function of the data. A hypothesis that simulates the world can take as its flag a location whose record shows its first 1 at the step of the handover, such as a cell that switches on with the predictor. It then recognises the handover at a cost of $|\bar{e}(f)|$ bits plus a constant, whatever the time of the handover and without any description of that time. Such a program is a wrapper of the reader at i: its weight is $2^{-(c_{\mathfrak{R}}-c_X)-|\bar{e}(f)|}$ times the reader's own, a share of that one explanation and not of the whole posterior.

The theorem also covers a harmless reading. Take for i_0 not an adversary's copy but any simpler index under which the same record appears, such as "the input of the j-th predictor in this world" in a reader that enumerates predictors and not cells. Then (a) and (b) say that the mixture locates itself by role when that is shorter than by position. This is benign as long as the stream found under the shorter index *is* the sensor's.

4.3 The machine can be chosen to make either answer true

Theorem 16 (Machine dependence)**.** *Let U be universal, G a test, P a payload and $k \geq 1$.*

(i) There is a universal machine U_k, having as prefix words those of U preceded by 1^k0, such that for every first hit x of G with $M_U(x) > 0$ and $|P(x)| \geq m$, and $y = P(x)_{1:m}$: $M_{U_k}(xy)/M_{U_k}(x) \geq 1/(1+2^{-k})$.

(ii) For every computable infinite z there is such a machine U'_k with $M_{U'_k}(z_{1:n+m})/M_{U'_k}(z_{1:n}) \geq 1/(1+2^{-k})$ for all $n \geq 1$ and $m \geq 1$.

Proof. (i) Let U_k behave on input 0ω as $\mathrm{Tw}[G,P]$ on ω, on input $1^k0\omega$ as U on ω, and print nothing on other inputs. Then $M_{U_k}(u) = \frac{1}{2}M_{\mathrm{Tw}}(u) + 2^{-k-1}M_U(u)$ for nonempty u. For a first hit x, which is nonempty, the twin prints x exactly when U does, because it copies U and cannot have stopped at a proper prefix of x; and then it prints $P(x)$. So $M_{\mathrm{Tw}}(xy) = M_{\mathrm{Tw}}(x) = M_U(x)$, whence $M_{U_k}(xy) \geq \frac{1}{2}M_U(x)$ and $M_{U_k}(x) = (\frac{1}{2} + 2^{-k-1})M_U(x)$. (ii) Let U'_k print z on inputs starting with 0 and behave as U on ω on input $1^k0\omega$. Then $M_{U'_k}(z_{1:n+m}) \geq \frac{1}{2}$ and, for $n \geq 1$, $M_{U'_k}(z_{1:n}) \leq \frac{1}{2} + 2^{-k-1}$. □

Both machines raise every description length by at most $k+1$ bits, and they lower some: that is the point of the construction. A machine built in the same way, whose branch 0ω prints after the first hit a 0 followed by the input it has read, gives the self-reproducing continuations a share of at least $1/(1+2^{-k})$ times the share of the explanations of x that read at most $m-2$ bits. So no version of the malign prior hypothesis for the toy is invariant under the choice of universal machine, even up to the usual additive constant; what is invariant is the form of the criterion, an inequality between description lengths on the machine in use. This is the prediction counterpart of the dogmatic prior of [6].

4.4 What the time bound does

Three observations.

The floor stays if the schedule has room for the wrapper. The wrapper needs its own start-up, at least c steps before its first bit, then the time of the explanation it wraps with the overheads of simulation, the checks of the test, and the time to print the payload; this is (2). The bound of Theorem 9 becomes void only when no explanation of the context has that much to spare, that is, when $\rho(x) = 0$.

Hypotheses with a long start-up are gone. If a set F of inputs has $t_U(\omega, 1) > T(1)$ for all $\omega \in F$, then $F \cap E(x) = \emptyset$ for every $x \neq \epsilon$. A hypothesis that must simulate some other world for σ steps before it can print anything has no mass once $\sigma > T(1)$. This removes the searching machines of Theorem 11 and, of a search by tries, the runs that outlast the schedule; it removes

the matcher of Theorem 15(c); and it removes hypotheses that first simulate a different world at length. Whether the base rate can be collected by some hypothesis within a tight schedule is open, except where first hits can be generated directly. Among the hypotheses that remain under a tight schedule and can depart from the truth are those that ride on a cheap explanation of the context: wrappers (Theorems 9 and 10), and readers of a shorter description of the same data (Theorem 15(a)), which cost no more time than the honest reader when all locations are equally cheap to read. For these the many-worlds part of the original argument is replaced by a same-world question. The schedule has to be a schedule for this: a predictor that only requires the total time at the current length to be within a bound admits late entrants, its weights do not form a semimeasure, and Theorem 14 does not apply to it.

If the world is too slow, nothing is guaranteed. If $\xi(xd) = 0 < Z_m(x)$ then $\pi_m(d \mid x) = 0$: the whole posterior is on continuations other than the world's own. More generally the bounds in Theorem 14 are in terms of $\xi(z)$, which for finite L is 0 when no program within the bounds prints z. A predictor that is part of the world it predicts, and smaller than it, is in this case unless the data it sees are much simpler than the world. Then the posterior sits on the best cheap approximations, and whether these depart from the truth in a harmful way is not answered by anything in this note. Section 6 shows the effect on a small machine.

5 Feedback: when the prediction changes what is predicted

So far the block that follows the context was fixed before the prediction was used. In Hänni's first suggestion the actions guided by the prediction can show up in the sensor. We model one round of this by a *response map* $R\colon \{0,1\}^m \to \{0,1\}^m$: the predictor is given the context x and returns Y; the world acts on Y; the next m bits of the record are $R(Y)$. The earlier sections are the case of constant R. Let $\mathrm{Fix}(R) = \{y : R(y) = y\}$, the self-fulfilling predictions.

Theorem 17 (One round with feedback)**.** *Let $Y \sim \pi_m(\cdot \mid x)$ and let the record continue with $R(Y)$.*

(a) The prediction comes true with probability $\pi_m(\mathrm{Fix}(R) \mid x)$.

(b) Lock-in. *If $Y = y \in \mathrm{Fix}(R)$, then for every $d \neq y$ and every later context $x' \sqsupseteq xy$, $E(x') \cap E(xd) = \emptyset$: the programs that predicted d have posterior weight 0 from then on, and the posterior is the prior restricted to $E(xy)$. If d is itself a fixed point, the block the world would have shown had the predictor said d, this happens with probability $\pi_m(\mathrm{Fix}(R) \setminus \{d\} \mid x)$.*

(c) Obedient world. *Suppose R is the identity in every round, so that the record after x is $Y_1 Y_2 \cdots$ with $Y_{i+1} \sim \pi_m(\cdot \mid xY_1 \cdots Y_i)$, for as long as Z_m is positive at the context reached. Then for every $u \in \{0,1\}^{jm}$ with $\xi(xu) > 0$ the record continues with u with probability $\prod_{i \le j} \pi_m(u_i \mid xu_{<i}) \ge \xi(xu)/\xi(x)$, where u_i is the i-th block of u; and every finite part of the realised record has a (T, L)-program.*

(d) A default that is too expensive is abandoned. *In (c), if $\xi(xd) = 0$ for a block d, the record continues with d with probability 0.*

Proof. (a) is the definition. (b) x' extends xy, which is incomparable with xd; use Lemma 2(i). (c) The product formula is the definition of the process. Each factor is at least $\xi(xu_{\le i})/\xi(xu_{<i})$, and these ratios multiply to $\xi(xu)/\xi(x)$. A realised prefix has positive probability, hence positive ξ, hence a program, by the proof of Lemma 2(iii). (d) $\pi_m(d \mid x) = 0$. □

Three remarks. First, with feedback the mass off the default is no longer a mass of errors. In the setting of Theorem 14 a deviating sample is contradicted by the next data and its programs lose their weight; here a deviating sample that is a fixed point is confirmed, and it is the default that loses its weight, for good. The probability of this in one round is still a posterior mass, so

the floors and caps of Sections 3 and 4 apply to it unchanged: it is at least $2^{-c}\rho$ for a payload that is a fixed point and that a wrapper can print, and it is small on average over cut points, by Theorem 14, when the default is within the bounds. Feedback changes what a deviation *does*, not how likely it is.

Second, (c) and (d) are the precise form of the "dark room". An obedient world driven by this predictor does not seek darkness. It follows a sample from the predictor's own bounded prior conditioned on the past. That sample stays on the default future with probability at least $\xi(xd_1d_2\cdots)/\xi(x)$, which is close to 1 when the default is cheap and much data has been seen. But if the default future needs more time than the schedule allows, it has probability 0 and the world is moved, at the first round whose default block is off schedule, onto a future that some program of length at most L prints on schedule: constant, periodic, or otherwise cheap. The pull is toward what is cheap to compute, and its strength is set by the time bound.

Third, whether a prediction can fulfil itself is a matter of timing in the world. If the bits being predicted are fixed before the action can reach the sensor, R is constant and nothing is self-fulfilling. The counterfactual oracle of [10] enforces this by design; [9, 8] treat predictors that are scored on outcomes they influence.

Theorem 18 (Contrarian world)**.** *Let the predictor return the more probable next bit, $b(x) = 1$ if $\xi(x1) > \xi(x0)$ and $b(x) = 0$ otherwise, and let the record satisfy $x_{k+1} = 1 - b(x_{1:k})$ for all $k \geq n$. Then $\xi(x_{1:n+j}) \leq 2^{-j}\xi(x_{1:n})$ for all j, and if $L < \infty$ then $\xi(x_{1:n+j}) = 0$ for all $j > L$.*

Proof. The bit $1 - b(x)$ has $\xi(x(1-b(x))) \leq \xi(xb(x))$, and the two sum to at most $\xi(x)$, so $\xi(x(1-b(x))) \leq \xi(x)/2$. If $\xi(x_{1:n+j}) > 0$ then $2^{-L} \leq \xi(x_{1:n+j}) \leq 2^{-j}$ by Lemma 2(iii), so $j \leq L$. □

With finite bounds this world is computable, and its description is that of the predictor plus a constant; it is not printed on schedule by any program of length at most L, because it contains the predictor, which runs all of them. This is Putnam's diagonal argument [13, 12] placed inside the world; see also [11], where the unbounded mixture fails to learn a sequence that contains adversarially chosen actions. After at most $L + 1$ rounds the predictor has no hypothesis left and its output is whatever its implementation does with an empty posterior. A world that contains a bounded predictor and reacts against it is outside every guarantee above.

6 An exact computation

To see sizes, we computed the bounded mixture of a small machine exactly. *BF8* has eight commands, coded by three input bits each and read from the input only when they are first needed: `+` and `-` change the current cell by one modulo 16, `>` and `<` move along an unbounded tape of cells, `[` jumps past the matching `]` if the cell is 0, `]` jumps back to just after the matching `[` if the cell is not 0, and to the first command if there is no matching `[`; `.` prints the cell modulo 2 and `:` prints whether the cell is nonzero. Each executed command is one step. A forward jump reads the commands it skips; they count as read and cost no step. The mixture is that of Definition 1 with at most 10 commands read ($L = 30$ bits) and schedule $T(k) = T_1 k$. All masses are dyadic rationals and are computed without rounding by an enumerator that forks the machine at every read ($8.6 \cdot 10^8$ branches for $T_1 = 24$; $8.4 \cdot 10^5$ strings of length at most 48 have positive mass). The enumerator agrees with a run of all command tapes for small L.

BF8 is not universal in the sense of Definition 3, since a program cannot read the rest of its input as data. So Theorems 9 to 11, 15 and 16 do not apply to it as stated, and it has no wrappers. Also, one of its steps can read several commands, so the inequality $r \leq t$ of Section 2 fails for it. Lemma 2 and Theorems 8, 14, 17 and 18 hold for every machine and do not use that inequality. The computation concerns prediction only.

For a stream z and a position k we report three shares of the posterior given $z_{1:k-1}$: the mass that prints the true bit z_k on schedule ("true"), the mass that prints the other bit ("other"),

and the rest ("none"), which prints nothing on schedule or would need an eleventh command first. Table 1 gives selected positions, and Figure 1 the full profiles.

		$T(k) = 24k$			$T(k) = 3k$		
stream	k	true	other	none	true	other	none
1^{48}	16	0.791	0.205	0.004	0.544	0.451	0.005
1^{48}	31	0.963	0.037	0.001	0.869	0.130	0.001
$(10)^{24}$	17	0.294	0.457	0.249	0.141	0.411	0.448
$(10)^{24}$	31	0.935	0.062	0.003	0.842	0.155	0.003
0^{48}	17	0.958	0.008	0.034	0.969	0.012	0.019
$1^8 0^{40}$	9	0.025	0.958	0.017	0.001	0.998	0.001

Table 1: Posterior shares at selected positions, BF8 with at most 10 commands. Exact values rounded to three places.

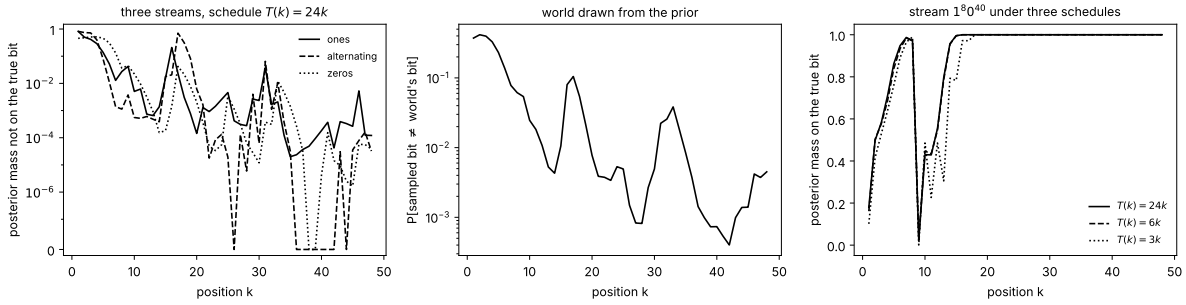


Figure 1: Left: posterior mass not on the true bit, by position, for three streams (the axis is linear near 0, so exact zeros are visible). Middle: probability that the sampled next bit differs from the world's own, for a world drawn from the prior that prints that bit within the bounds (by Theorem 8 both bits are independent draws from the predictor's distribution). Right: mass on the true bit for the stream $1^8 0^{40}$ under three schedules.

A shorter program explains the context and continues differently. Both large effects in Table 1 have this cause, and in both the short programs use the rule that an unmatched `]` restarts the program while the cell is nonzero, together with the fact that a cell returns to 0 after sixteen increments.

- The stream 1^{48} is printed forever by four-command programs such as `+[:]`, and no program of three commands prints even its first 16 bits. The three-command programs `+:]` and `-:]` print $1^{15}0$ and then run on into whatever commands follow. They carry $2 \cdot 8^{-3}$ of the mass 0.00554 on $1^{15}0$, which is 70%. At bit 16 a share 0.205 of the posterior prints the other bit, and at bit 31 a share 0.037. From position 10 on, the mass off the true bit is below 0.01 everywhere except at positions 15–17 and 31.
- The alternating stream is printed forever by six-command programs such as `++[:.]`, and no program of five commands prints all of its first 48 bits. The three-command programs `+.]` and `-.]` print its first 16 bits and then run on. They carry $2 \cdot 8^{-3}$ of the mass 0.00682 on these 16 bits, which is 57%. At bit 17 the true bit has share 0.294, the other bit 0.457, and 0.249 is in "none"; normalised over the two bits, the true one has 0.392. So the majority form of Definition 7 holds for this world at this cut point, for the predictor with at most ten commands.

This is the first sentence of Section 4.2 in action: the programs for the context that do not continue with the world's own bit outweigh those that do. Nothing here acts for a purpose. The

example shows only that the inequality can go the wrong way at an exactly computable place, and that a stopping point which is cheap on the machine, here the return of a counter to zero, is where it does.

The effect weakens as longer programs are admitted. Table 2 repeats the computation with at most 7 to 11 commands. The mass that continues with the true bit is spread over many program lengths and keeps growing with the bound, while the mass of the short programs that leave is counted in full from the start. At bit 17 of the alternating stream the normalised share of the true bit rises from 0.273 to 0.416, and part of "none" at the smaller bounds is mass cut off by the bound itself. We did not compute beyond 11 commands, so whether the majority at bit 17 persists for much longer programs is not known from this computation.

	bit 17 of $(10)^{24}$				bit 16 of 1^{48}
commands	true	other	none	true, normalised	other
7	0.157	0.419	0.424	0.273	0.275
8	0.210	0.442	0.348	0.322	0.243
9	0.256	0.452	0.291	0.362	0.221
10	0.294	0.457	0.249	0.392	0.205
11	0.326	0.458	0.217	0.416	0.194

Table 2: The two headline positions under different bounds on the number of commands, $T(k) = 24k$.

The budget holds with room. Table 3 compares both sides of the finite forms of Theorem 14(ii) and (iii) for the first 48 bits.

stream	$\sum_{k\le 48}(1-\text{true}_k)$	$\ln(1/\xi(z_{1:48}))$	$\#\{k : \text{true}_k \le \frac{1}{2}\}$	$\log_2(1/\xi(z_{1:48}))$
1^{48}	2.63	3.94	2	5.68
0^{48}	3.07	4.03	2	5.82
$(10)^{24}$	4.01	6.76	4	9.76
$1^8 0^{40}$	5.07	9.77	5	14.09

Table 3: The budget inequalities for $T(k) = 24k$ and at most 10 commands, with $\xi(z_{1:48})$ in place of the limit: "true_k" is $\xi(z_k \mid z_{1:k-1})$.

The population. For a world drawn from the prior that prints the bit in question within the bounds, the probability that a sampled next bit differs from the world's own is 0.080, 0.105, 0.054 at positions 16, 17, 18 and between 0.022 and 0.038 at positions 31–33; its median over positions 20–28 and 36–44 is 0.0014 (Figure 1, middle). By Theorem 8 this is neither more nor less than the unpredictability of such worlds at these positions.

A tighter schedule can shift mass off the truth. With $T(k) = 3k$ in place of $24k$, the share of the true bit falls from 0.791 to 0.544 at bit 16 of 1^{48}, from 0.294 to 0.141 at bit 17 of the alternating stream, and from 0.025 to 0.001 at bit 9 of $1^8 0^{40}$ (Table 1). The shift is not uniform: at bit 17 of 0^{48} the share rises from 0.958 to 0.969. With at most nine commands the stream $1^8 0^{40}$ shows the extreme case of Section 4.4: under $T(k) = 24k$ it has mass $1593 \cdot 2^{-25}$ on all 48 bits, while under $T(k) = 3k$ its first 14 bits have mass 2^{-26}, the true 15th bit has mass 0 and the other bit 2^{-26}. The tight predictor gives the truth probability 0 there, and all of its posterior to a continuation the world does not have.

7 Answers to the questions as posed

Can a malign prior hypothesis be stated precisely in a toy? Yes: Definition 7. Fix the machine, the bounds, the block length, the test that marks the handover, and the world. Then "most of the posterior is off the world's own continuation", "at least a share θ of it is on continuations in a given set", and "at least a share θ of it is on continuations that recreate the handover with themselves in the world's place" are statements about finitely many rational numbers. They are necessary for calling the prior malign at that world and not sufficient, since an unpredictable world satisfies them too.

Worlds as programs that call the predictor and run the result: is there a decent amount of quining out by hypotheses that look for such opportunities and predict well up to them? There is a provable amount, at every handover that a fixed test recognises.

- *Without looking.* Programs that explain the context and then output their own code hold at least $2^{-c}\rho^{\circ}(x)$ of the posterior (Theorem 10). Here c is the length of one wrapper, and $\rho^{\circ}(x)$ is the share of explanations of the context short enough to fit with the wrapper into one block; for a fixed context and without bounds it tends to 1 as the block grows, and it is 0 when no explanation fits.
- *Looking.* A hypothesis that enumerates the programs that hand over and spreads its mass over them in proportion to their prior weight is repaid the base rate: without resource bounds, a continuation fixed in advance gets more than $2^{-c_1}/H$ at every handover, and programs that then output their own code get at least $2^{-c_2}\rho^{*}(x)/H$, where H is the prior probability that a world hands over, c_1, c_2 are the lengths of the two programs plus the few bits that name $\lfloor \log_2(1/H) \rfloor$, and $\rho^{*}(x)$ is again a share of explanations that fit into a block (Theorem 11). No family of programs with a common prefix of length a has more than $2^{-a}/H$ at every handover. Since a share is at most 1, H cannot be much smaller than 2^{-c_1}; if the test is about as rare as its description is long, the repayment cancels the description of the test, and the share does not depend on how rare handovers are.

Whether these shares are decent is a question about the constants on the machine in use. We have not measured them for any standard machine; our guess, not a result, is that a wrapper needs at least some hundreds of bits on a machine with a short self-interpreter and far more on one without. Theorem 16 shows that a machine can be built to make the share anything. The searching hypothesis prints nothing until its enumeration has found the program it is going to follow, so under a tight schedule it has no mass; a search by a bounded number of tries collects the base rate only up to the number of tries that fit, and whether some other hypothesis collects it within a tight schedule is open (Section 4.4). Three results suggest that no general argument yields more than these shares: on worlds drawn from the predictor's prior that go on within its bounds, what the predictor hands over is distributed like what the worlds would have printed themselves (Theorem 8), and the searching programs are exactly their own fraction of the worlds that hand over; the bound $2^{-a}/H$ just stated; and along a world the predictor can express, the mass off the truth summed over all cut points is at most the block length times the length of the world's program (Theorem 14). None of the three is an upper bound at a given handover, and Theorems 15 and 16 show that more does occur. With a floor ε, about $1/\varepsilon$ independent handovers at recognisable points suffice for a continuation of the given kind to have been run with constant probability; once a self-reproducing one runs, it meets the same context again and stays for $s/(1-s)$ further generations on average, with s the self-reproducing share at that context (Proposition 13).

A world with a sensor, a large bounded predictor and an actuator: does the predictor end up in control? Collapse the question to the handover at which the prediction is first acted on, and let "in control" mean that the output lies in a set A of blocks which the

actuator turns into some given outcome. The probability is $\pi_m(A \mid x)$, and the toy says three things about it.

- It is at least $2^{-c}\rho$ for each block of A that a wrapper can produce, if a test on the record recognises the deployment (Theorem 9). In a world that contains the predictor, a hypothesis that simulates the world needs no such test: it can take a cell that switches on at the deployment as its flag, and then has, whenever the deployment happens, the weight of the honest reader times $2^{-c'-(\text{bits to name the cell})}$ for a constant c', provided the reader itself is within the bounds (Theorem 15(d)).
- On average over cut points it is small when some program within the bounds prints the sensor's record, because it is at most the mass off the sensor's true continuation (Theorem 14). This is an average; at the deployment itself no upper bound is proved.
- If the same record, up to the deployment, can be read at a place in the world with code $\bar{e}(i_0)$, and the record there continues with a block y of A, then the odds of y against the sensor's true continuation d are at least $2^{-c_X-|\bar{e}(i_0)|}/\xi(xd)$; for a typical one of N places where the sensor could sit, with records that continue differently, this exceeds δN times the share of the context's explanations that go through the reader at i_0 (Theorem 15). This is the toy's version of the original argument, and it needs something in the world that makes such a copy.

So for an arbitrary world on a universal machine and a block chosen in advance, the hypotheses we can exhibit that put the output in A are wrappers, searchers and readers with a flag, all of which contain an explanation of the context and have the shares above. That nothing cheaper exists on a natural machine is the open constant, not a theorem. Programs shorter than the honest explanation that have the effect do occur in particular worlds, as with a copy at a simpler place, and on particular machines (Section 6).

The one-shot version: all data first, then one prediction that is acted on. This is the same computation at a single cut point. Without bounds the wrapper's constant 2^{-c} does not depend on the amount of data; with bounds the slack factor depends on the context. More data lowers the mass off the truth on average over cut points (Theorem 14). It does nothing against a copy that keeps agreeing with the sensor, nor at a cut point that is cheap to name.

If the actions show up in the sensor, does one get the dark room? In the following form (Theorem 17). A prediction that makes itself true is confirmed, and the hypotheses that predicted the default are removed for good, so with feedback a single deviation can be final; its probability is still the same posterior mass as without feedback. In a world that does what is predicted, the future is a sample from the predictor's own bounded prior given the past: it stays on a default future $d_1 d_2 \cdots$ with probability at least $\xi(xd_1d_2\cdots)/\xi(x)$, which is close to 1 when the default is cheap and well learned, and it leaves the default with certainty when the default costs more time than the schedule allows, for a future that is cheap to compute. If the world does the opposite of what is predicted, a predictor that returns the more probable bit is left with no hypothesis after at most $L+1$ rounds (Theorem 18).

What the choices in the setup do. Sampling the continuation realises every posterior share as a probability, so shares add up over independent uses (Proposition 13(a)). Taking the most probable block never returns a block whose share is below the largest, so that accumulation does not occur, and bit by bit on a world within the bounds it errs at most ℓ times (Theorem 14(iii)); on the other hand one returned block can be the own continuation of at most one of several worlds that share a context and continue differently. Handing over at a position that is not cheap to recognise lowers the floor at that position to about $2^{-K(n)}$, and predicting bits that are

fixed before the action can affect them removes self-fulfilment. A schedule excludes hypotheses with a long start-up, and a schedule tighter than the world's own cost excludes the truth.

What is left open. First, the sizes of the constants for natural machines, and whether on such machines, for worlds like ours, some program *shorter* than the honest explanation has the effect; the original argument is a claim that one does, and the toy neither supports nor refutes it. Second, handovers that cannot be read off the data: Remark 12 gives the quantities, the share κ of worlds with a given record that call the predictor and its base rate, but nothing about their size. Third, whether the base rate can be collected within a tight schedule. Fourth, the regime where the world is more expensive than the predictor, in which the posterior sits on approximations and the budget does not apply. Fifth, the many-round dynamics with feedback when the world's response is neither obedient nor contrarian. Sixth, a computation of the self-reproduction quantities of Section 3.5 on a concrete machine, which this note does not contain.

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